

JARRELL I. REYES

43 Narra Street, Talanay A, Batasan Hills, Quezon City, Philippines
Email: jrrllrey17@gmail.com | Phone: 09564330929

PROFESSIONAL SUMMARY

High-performing Full-Stack and Mobile Software Engineer specializing in scalable enterprise web architectures and secure, hardware-integrated mobile applications. Proven expertise designing offline-first synchronization protocols, hardening transactional security systems, and maintaining high-availability production environments. Expert in translating complex corporate operational workflows into robust, automated software solutions.

EDUCATION

Polytechnic University of the Philippines
Bachelor of Science in Information Technology

Graduation: 2021 – 2025

EXPERIENCE

Freelance Full-Stack & Mobile Developer
Self-Employed Software Engineer

February 2026 – Present

- **Ecosystem Architecture:** Architected, deployed, and managed multi-platform application suites for enterprise clients, encompassing administrative web dashboards, offline-first field operation applications, and customer portals.
- **Data Integration & Synchronizations:** Designed and configured low-latency, resilient data sync protocols between local SQLite databases on mobile clients and centralized MVC backends, ensuring absolute transactional integrity.
- **Stakeholder Collaboration:** Liaised directly with business owners to gather requirements, perform business analysis, and systematically transition manual cash and service cycles into automated software pipelines.
- **Transactional Security & Integrity:** Hardened transactional, financial, and access pathways by implementing robust Webhook processors, Role-Based Access Controls (RBAC), and unique sequencing pipelines to mitigate security gaps.

Telus
Technical Support Engineer (Google RCS Ecosystem)

July 2025 – Present

- Managed cloud-based infrastructure onboarding protocols for Rich Communication Services (RCS) to facilitate digital enterprise communication frameworks.
- Liaised strategically between Google engineering teams and enterprise telecom clients to troubleshoot and streamline carrier deployment API cycles.

Polytechnic University of the Philippines
Intern (Project Lead / Business Analyst / Lead Full-Stack Dev)

Feb 2025 – June 2025

- Spearheaded system architecture, scope boundary analysis, database schema modeling, and full-stack development for a multi-user academic administrative portal.

Software Research Group
Project Manager & Front-End Developer

Oct 2024 – Feb 2025

- Championed Agile software development ceremonies and drove the core frontend lifecycle for highly responsive system interfaces built with Flutter and Dart.

Capstone Project
Project Manager & Lead UI Developer

Aug 2024 – Feb 2025

- Directed UI/UX specifications, modular task delegation structures, and requirements modeling, translating operational user friction points into developer-ready backlogs.

SELECTED PROJECTS & ENTERPRISE SYSTEMS

Tubig Pilipinas Admin System | GT Industries

Enterprise Web Application

Core Technologies: PHP, CodeIgniter MVC, MySQL, JavaScript, jQuery, Git, XAMPP, Excel (.xlsx)

- **Financial Ledger Engine & Billing Restructuring:** Re-engineered the transactional Statement of Account (SOA) rendering workflow to enforce strict, sequential compound billing calculations (Basic Charges → VAT → Arrears → Penalties), eliminating critical calculation anomalies.
- **Targeted Communication Channels:** Enhanced client compliance by embedding dynamic, contextual warning banners and generating high-accuracy PayMongo-integrated QR codes onto generated bills to facilitate mobile self-service payments.
- **Isolated Cashier Transactions (SI Series):** Designed a thread-safe transaction-sequencing pipeline that assigns unique, isolated numeric ranges to cashier systems. Built reactive background listeners to notify operators near series exhaustion, ensuring zero-collision receipt auditing.
- **Audit-Grade Excel Synchronization:** Built a robust, memory-efficient excel data parsing engine (.xlsx) featuring multi-column uniqueness constraints to support high-volume transaction uploads, eliminating raw DB insertions and manual reconciliation.
- **System Integrity & Production Security:** Mitigated production-critical HTTP 500 runtime errors through strict MVC model restructuring and database type constraints. Hardened security by deploying global RBAC authorization middleware in the Base Controller to systematically prevent client privilege escalation.

Meter Reader App | GT Industries

Field Operations Mobile Application

Core Technologies: Flutter, Dart, SQLite, Local DB, REST APIs, Camera Photo Capture

- **Mobile Billing Parity & Document Generation:** Standardized the mobile-printed SOA layout with the primary database schemas, restructuring the fee order to prioritize compound calculations (Basic → VAT → Arrears → Penalties). Eradicated legacy ledger discrepancies, configured client warning notifications, and integrated PayMongo-based e-payment QR triggers.
- **Offline-First Caching & Mutability:** Architected local SQLite tables enabling field teams to execute offline modifications on queued reads. Implemented mandatory photo-proof validations that flag updated entries for admin review before synchronization.
- **Asynchronous Customer Care Hub:** Developed an offline-first support ticket logging module allowing asynchronous client service registrations, local file caching of captured images, and automated synchronization of issues (leaks, defects, adjustments) on connectivity restore.
- **Low-Level Hardware & Serial Search:** Integrated ESC/POS thermal printing configurations to dynamically render billing files on-site, standardizing reference IDs to a clean, 7-digit incremental format. Designed rapid offline search functions using local serial number arrays.
- **State Handling for Operational Anomalies:** Engineered partial reading support structures to safely isolate replaced meters, managing distinct transitions and sequential device registers without corrupting parent consumer profiles.

Tubig User App | GT Industries

Customer Self-Service Mobile Application

Core Technologies: Flutter, Dart, PayMongo SDK, REST APIs, iOS & Android

- **Unified Customer Ledger Parity:** Refactored the customer-facing Statement of Account (SOA) data layer to mirror standardized multi-tier fee structures. Removed outdated account balance lines to guarantee absolute transaction clarity.
- **Frictionless Mobile Settlements:** Integrated a PayMongo payment module directly within digital statements, allowing concessionaires to execute secure, real-time bill payments directly from their devices to significantly reduce physical foot traffic.
- **Contextual Notices & Safety Prompts:** Standardized in-app communication headers with automated payment reminders and strict cut-off warnings, minimizing account maintenance overhead.

Mosang's Budbod

Full-Stack Food Logistics & Ordering System

Core Technologies: Google Firebase (Auth, Firestore, Cloud Storage), PayMongo Webhook APIs

- **System Migration & Reverse Engineering:** Reverse-engineered an undocumented, legacy database to systematically migrate data assets to Google Firebase without experiencing live-production downtime.
- **Third-Party Logistics Automation:** Implemented structural dispatch and courier integrations using third-party transit APIs to generate dynamic, distance-based pricing models and live order lifecycle tracking.
- **Automated Ledger Synchronization:** Configured multi-direction webhook receivers for PayMongo processing, ensuring immediate transactional accounting updates and system status changes on payment confirmation.

Other Projects & Academic Systems

Core Technologies:

- **QR-Attendance System** – Designed and implemented a full-stack automated attendance tracker in Python/Django utilizing PostgreSQL and custom high-density QR scanning. (2025)
- **Alumni Tracker System** – Created a Django-based web portal with Bootstrap 5 and Jira to manage coordinate registrations and profiles for university alumni. (2025)
- **GFMIC Membership Platform & Mobile App** – Built responsive Django and Flutter interfaces backed by RESTful APIs to handle GFMIC member registration databases. (2025)
- **PUPQC-CRGS (Cashier & Financial System)** – Refactored Laravel transaction processing workflows to optimize PUP cashier desks and secure cash reconciliation sheets. (2025)
- **DALAY Barangay Automation** – Developed a PHP/Laravel citizen management system with Bootstrap for local Philippine government profiles, Blotter cases, and legal logs. (2024 – 2025)
- **Patient Management System** – Built a PHP/Laravel hospital platform for tracking medical records, clinic queues, assessing emergency admissions, and appointments. (2024 – 2025)
- **Bus Reservation Mobile App** – Developed a mobile travel portal with ticket purchases, custom seat matrices, and map APIs using Flutterflow and Firebase. (2023 – 2024)
- **Al Sarap Cooking Guide App** – Engineered a Flutterflow mobile application running custom sorting algorithms to recommend recipes based on dynamic kitchen pantry counts. (2023 – 2024)
- **SmartStudy Student Suite** – Designed a specialized student suite in Flutterflow with local study timers, mark-down note renderers, and focus analytics. (2023 – 2024)
- **Algoritees E-Commerce Web App** – Built a PHP/Laravel online storefront featuring digital checkout flows, merchandise inventory ledgers, and catalog managers. (2024)
- **Payroll Management System** – Created an automated PHP/Laravel employee payroll pipeline calculated around regional tax rules, deductions, and financial exports. (2022 – 2023)

CORE TECHNICAL SKILLS

- **Languages & Frameworks:** Python (Django), PHP (Laravel, CodeIgniter), JavaScript (ES6+), TypeScript, Next.js, HTML5, CSS3, Sass, Bootstrap, Dart (Flutter), React Native.
- **Databases, Cloud & Storage:** PostgreSQL, MySQL, Google Firebase (Firestore, Authentication, Cloud Storage), SQLite, AsyncStorage.
- **Integrations & Protocols:** RESTful APIs, Webhooks, Pusher WebSockets, Firebase Cloud Messaging (FCM), Notifee, ESC/POS Bluetooth Printing, PayMongo, Angkas Padala API.
- **Tools, Platforms & DevOps:** Git, GitHub, Jira, Agile/Scrum, Android Studio, Gradle, XAMPP, MS Excel Data Pipelines.
- **Core Competencies:** System Architecture Design, Secure Database Engineering, Offline-First Synchronization, High-Performance Mobile Layouts, Role-Based Access Control (RBAC), Technical Requirements Modeling.

CERTIFICATIONS

- Responsive Web Design (2025)
- Javascript Algorithm and Data Structures (2025)
- Google Project Initiation: Starting a Successful Project (2024)
- Google Project Planning: Putting it all Together (2024)
- Foundation of Project Management (2024)
- Business Analysis & Process Management (2024)
- Fortinet - Getting Started in Cybersecurity 2.0 (2024)
- mabl skill Certification: Foundations (2024)
- Nemesis Prudente International Research Conference (2024)
- Design App Completion (2024)
- Programming for Intermediate Users Using Python (2022)

LICENSE / ELIGIBILITY

Civil Service Eligibility - Professional Level

Issued by the Civil Service Commission (CSC), Republic of the Philippines

PROFESSIONAL REFERENCES

Ms. Demelyn E. Monzon, PhD.

Head, Office of Academic Programs
Polytechnic University of the Philippines
Quezon City Campus
Contact: 09453202215
Email: demonzon@pup.edu.ph

Ms. Rosicar E. Escobar, PhD. DIT

Associate Professorial Lecturer 5, PUP
Enterprise Solutions Head, NGCP
Polytechnic University of the Philippines
Quezon City Campus
Contact: 09175980384
Email: joroseescober@gmail.com